Spam mail — Naive Bayes

kazandiniz → spam

indirim → spam

proje → ham

merhaba → ham

kelimeler birbirine bağlı değilse

$$P(x_1 \mid x_2) = \frac{P(x_1, x_2)}{P(x_2)} \overset{1}{=} \frac{P(x_1)\,P(x_2)}{P(x_2)} = P(x_1)$$

$$P(\text{spam} \mid x_1, \dots, x_N) \overset{?}{\geq} P(\text{ham} \mid x_1, \dots, x_N)$$

$$P(y \mid x_1, \dots, x_N) = \frac{\overbrace{P(x_1, \dots, x_N \mid y)}^{\text{likelihood}}\; \overbrace{P(y)}^{\text{prior distribution}}}{P(x_1, \dots, x_N)}$$

$$\underbrace{P(x_1, \dots, x_N, y)}_{\text{joint distribution}} = P(x_1 \mid x_2, \dots, x_N, y)\, P(x_2, \dots, x_N, y)$$

$$= P(x_1 \mid x_2, \dots, x_N, y)\, P(x_2 \mid x_3, \dots, x_N, y)\, P(x_3, \dots, x_N, y)$$

$$= P(x_1 \mid x_2, \dots, x_N, y)\, P(x_2 \mid x_3, \dots, x_N, y) \cdots P(x_N \mid y)$$

kelimeler bağımsızsa $\Rightarrow$

$$= P(x_1 \mid y)\, P(x_2 \mid y) \cdots P(x_N \mid y)$$

$$\frac{P(x_1, \dots, x_N \mid \text{spam})\; P(\text{spam})}{P(x_1, \dots, x_N \mid \text{ham})\; P(\text{ham})} \geq 1$$

↑ Bu ödev olacak

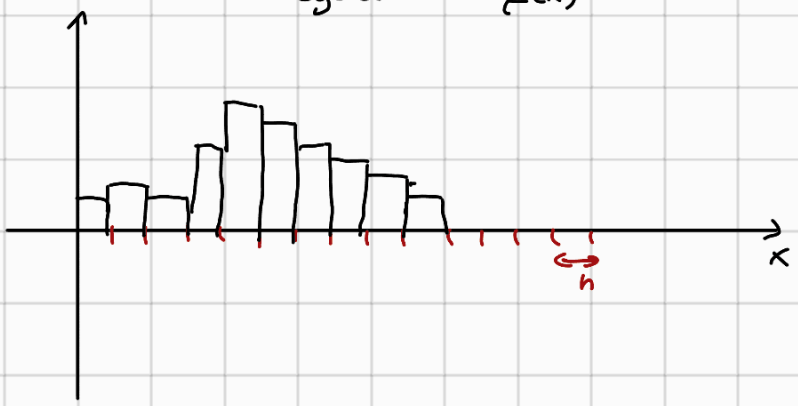
* $P(C_i)$
* $P(x_1 \mid C_i) = \frac{P(x_1, C_i)}{P(C_i)}$

Burda bernoulli dağılımı kullandık, gauss da kullanılabilirdi.

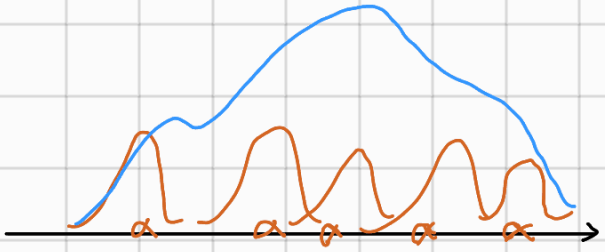
sklearn naive bayes sınıflandırıcı

Kitapta non parametrik kısımlara girecek

histogram $f(x)$



olasılık dağılımı için kullanılabir o zaman $\int f(x) = 1$ olmalıdır.

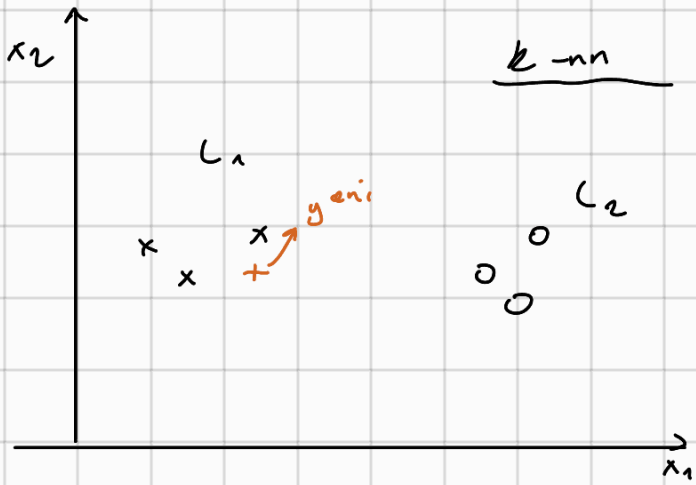


$$\frac{1}{C}\sum_{i=1}^{N} K\left(\frac{x - x_i}{h}\right) = f(x)$$

→ varyansa göre varyans büyükse h küçülsün

Kernel Approximation fonksiyonlar birbirinin aynısı

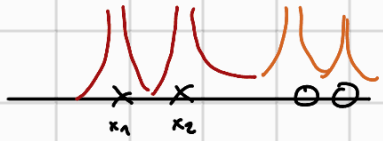
window functions



Burda bütün verileri tutmamız lazım k-nn de

distance metriği ile çözersek

$d = |x - x_i|$



kernel fonksiyonu $k = \frac{1}{|x - x_i|}$

1nn için $\max_i \frac{1}{|x - x_i|}$

2-nn için en büyük ikinci için al

$\min_i (k_i)$

make blobs

kernel density

Pseudo code seuler: